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Compatibilidad $\mathcal{C}^\infty$ entre cartas

Definición 1 ($\mathcal{C}^\infty$-compatibilidad). Sean (U_1, ψ_1), (U_2, ψ_2) dos cartas de dimensión n en X, son $\mathcal{C}^\infty$-compatibles $\iff$

(i) $\psi_2 \circ \psi_1^{-1}\big|_{\psi_1(U_1 \cap U_2)} : \psi_1 (U_1 \cap U_2) \longrightarrow \psi_2 (U_1 \cap U_2)$ es $\mathcal{C}^\infty$.

(ii) $\psi_1 \circ \psi_2^{-1}\big|_{\psi_2(U_1 \cap U_2)} : \psi_2 (U_1 \cap U_2) \longrightarrow \psi_1 (U_1 \cap U_2)$ es $\mathcal{C}^\infty$.

Ejercicio 1. Prueba que la $\mathcal{C}^\infty$-compatibilidad es una relación de equivalencia.

Referenciado en

- `atlas-diferenciable`
- `lem-estructuras-diferenciables-iguales`
- `teo-existencia-unicidad-estructura-diferenciable`
- `prop-atlas-unicarta-imp-diferenciable`
- `esp-proyectivo-real`
- `prop-estructura-diferenciable-inducida-homeomorfismo`

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